ECET 250E-Linear Circuit 1
Problemset 6-Chapter 7- First Order RC-RL Circuits

Name: _____ Marks: _____

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1. **7.2** Find the time constant for the *RC* circuit in **Fig. 7.82**.

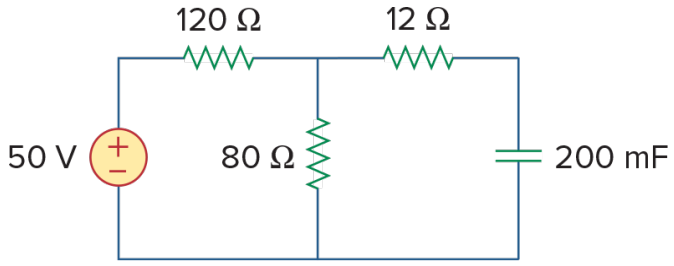


Figure 7.82 For **Prob. 7.2**.

2. **7.6** The switch in **Fig. 7.86** has been closed for a long time, and it opens at t=0. Find v(t) for t≥0.

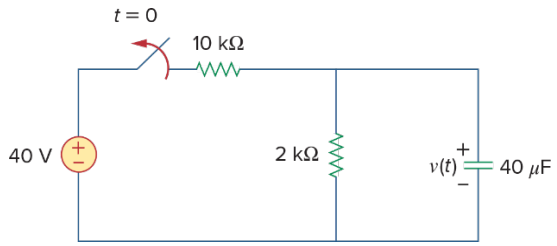


Figure 7.86 For **Prob. 7.6**.

3. 7.9 The switch in Fig. 7.89 opens at t=0. Find vo for t>0.

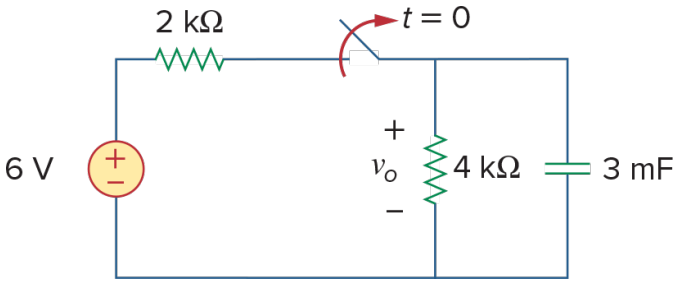


Figure 7.89 For Prob. 7.9.

4. 7.11 For the circuit in Fig. 7.91, find io for t>0.

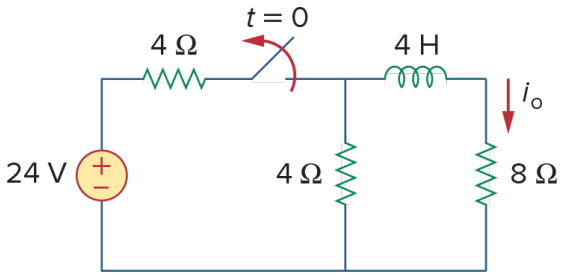


Figure 7.91 For Prob. 7.11.

5. 7.13. In the circuit of Fig. 7.93,

$$v(t) = 80e^{-10^3 t}\ \text{V}, \quad t > 0$$
$$i(t) = 5e^{-10^3 t}\ \text{mA}, \quad t > 0$$

(a)Find *R, L*, and τ.
(b)Calculate the energy dissipated in the resistance for 0<t<0.5 ms.

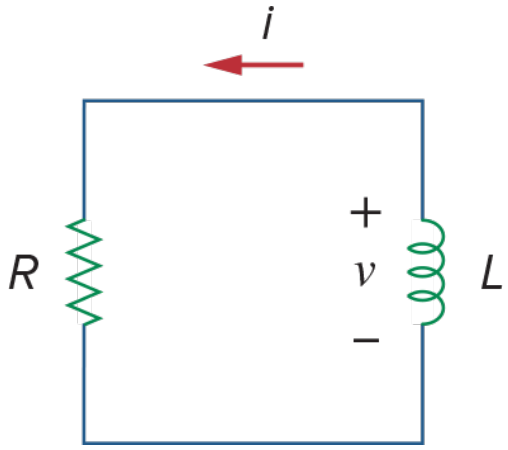


Figure 7.93 For Prob. 7.13.

6. **7.14** Calculate the time constant of the circuit in Fig. 7.94.

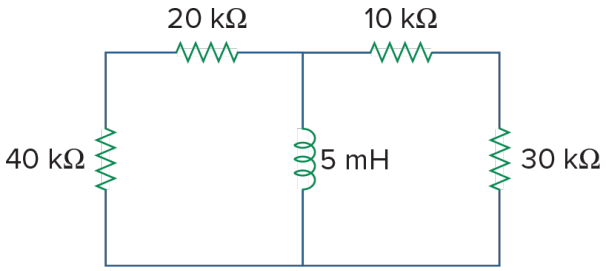
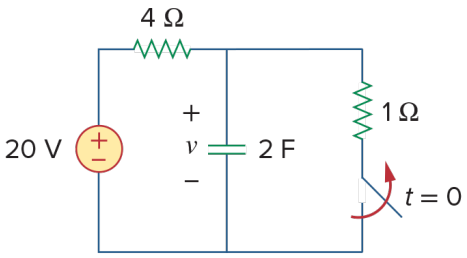
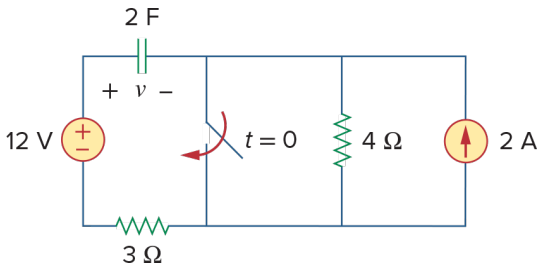


Figure 7.94 For Prob. 7.14.

7. **7.39** Calculate the capacitor voltage for t<0 and t>0 for each of the circuits in Fig. 7.106.



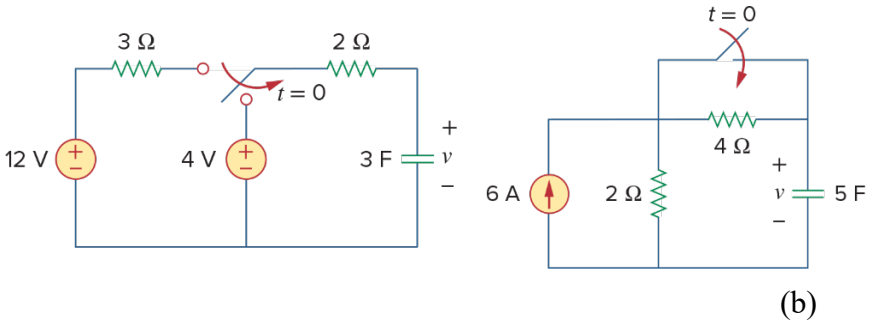
(a)



(b)

Figure 7.106 For Prob. 7.39.

8. 7.40 Find the capacitor voltage for t<0 and t>0 for each of the circuits in Fig. 7.107.



(a) (b)

Figure 7.107 For Prob. 7.40.

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

9. 7.42 (a)If the switch in **Fig. 7.109** has been open for a long time and is closed at t=0, find $v_o(t)$.

(b)Suppose that the switch has been closed for a long time and is opened at t=0. Find $v_o(t)$.

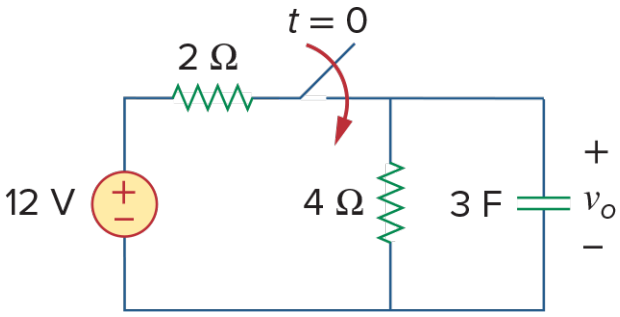


Figure 7.109 For **Prob. 7.42**.